

```

/* Project 2B - ADEX Creation */;
options validvarname=upcase nofmterr;

%let SDTMPATH = /home/u64124548/sasuser.v94/ADAM - Creation/SDTM_Data_Input;
%let OUTPUTPATH = /home/u64124548/sasuser.v94/ADAM - Creation/ADAM_Output_LogFiles;
%let VALPATH = /home/u64124548/sasuser.v94/ADAM - Creation/Validation_Data_ADAM;

libname adam "&OUTPUTPATH";

proc printto log="&OUTPUTPATH/ADEX.log" new;
run;

libname xda xport "&SDTMPATH/DA.xpt";

proc copy in=xda out=work;
run;

proc sort data=adam.adsl out=adsl;
  by USUBJID;
run;

proc sort data=da out=da_nodup nodupkey;
  by USUBJID DARFTDTC DADTC DATESTCD DAORRES;
run;

proc sql;
  create table dose_caps as
  select USUBJID,
         sum(DASTRESN) as CUMCAP
  from da_nodup
  where DASTRESN ne . and upcase(DATESTCD)='TAKENAMT'
  group by USUBJID;
quit;

proc sort data=dose_caps;
  by USUBJID;
run;

data ex_base;
  length
    STUDYID $8 USUBJID $21 SUBJID $5 SITEID $6 INVNAM $50 INVID $5
    AGEU $5 AGEGR1 $8 SEX $1 RACE $32 ETHNIC $22 COUNTRY $3
    TRT01P $10 TRT01A $10 TRTP $10
    REMISS $40
    ITTFL SAFFL EFFL CA125FL TRTDCFL COMPLFL STDDCFL SFUDCFL SFUFL $1;

  merge adsl(in=a
    keep=STUDYID USUBJID SUBJID SITEID INVNAM INVID AGE AGEU AGEGR1 AGEGR1N
    SEX RACE ETHNIC COUNTRY RANDDT TRT01P TRT01PN TRT01A TRT01AN
    REMISS REMISSN ITTFL SAFFL EFFL CA125FL TRTDCFL COMPLFL STDDCFL
    SFUDCFL SFUFL TRTSDT TRTEDT FPDUR)
    dose_caps;
  by USUBJID;

  if a;

```

```
TRTP = TRT01P;

if CUMCAP=. then CUMCAP=0;

if TRTSDT ne . and TRTEDT ne . then do;
    TXDUR = ((TRTEDT - TRTSDT) + 1) / 30.4375;
    TXDAYS = TRTEDT - TRTSDT + 1;
end;

CUMDOSE = (CUMCAP * 150) / 1000;

if TXDAYS > 0 then INTENS = (CUMCAP / TXDAYS) * 100;

format RANDDT TRTSDT TRTEDT is8601da.;
run;
```

```
data ex1;
    set ex_base;
    length PARAM $40 PARAMCD $8 DTYPE $40;
    PARAM='Duration (months) treatment received';
    PARAMCD='TXDUR';
    AVAL=TXDUR;
    DTYPE='DIFFERENCE';
run;
```

```
data ex2;
    set ex_base;
    length PARAM $40 PARAMCD $8 DTYPE $40;
    PARAM='Total Number of 150mg capsules taken';
    PARAMCD='CUMCAP';
    AVAL=CUMCAP;
    DTYPE='SUM';
run;
```

```
data ex3;
    set ex_base;
    length PARAM $40 PARAMCD $8 DTYPE $40;
    PARAM='Total Cumulative Dose (g)';
    PARAMCD='CUMDOSE';
    AVAL=CUMDOSE;
    DTYPE='SUM';
run;
```

```
data ex4;
    set ex_base;
    length PARAM $40 PARAMCD $8 DTYPE $40;
    PARAM='Dose Intensity (%)';
    PARAMCD='INTENS';
    AVAL=INTENS;
    DTYPE='PERCENTAGE';
run;
```

```
data adam.adex(label="Exposure Analysis");

    length
        STUDYID $8 USUBJID $21 SUBJID $5 SITEID $6 INVMAM $50 INVID $5
```

```

AGEU $5 AGEGR1 $8 SEX $1 RACE $32 ETHNIC $22 COUNTRY $3
TRT01P $10 TRT01A $10 TRTP $10
REMISS $40
ITTFL SAFFL EFFL CA125FL TRTDCFL COMPLFL STDDCFL SFUDCFL SFUFL $1
PARAM $40 PARAMCD $8 DTYPE $40;

```

```
set ex1 ex2 ex3 ex4;
```

```
format RANDDT TRTSDT TRTEDT is8601da.;
```

```
label
```

```

STUDYID = "Study Identifier"
USUBJID = "Unique Subject Identifier"
SUBJID = "Subject Identifier for the Study"
SITEID = "Study Site Identifier"
INVNAM = "Investigator Name"
INVID = "Investigator Identifier"
AGE = "Age"
AGEU = "Age Units"
AGEGR1 = "Age Group 1 (Char)"
AGEGR1N = "Age Group 1 (Num)"
SEX = "Sex"
RACE = "Race"
ETHNIC = "Ethnicity"
COUNTRY = "Country"
RANDDT = "Randomization Date"
TRTP = "Planned Treatment (Record Level)"
TRT01P = "Planned Treatment"
TRT01PN = "Planned Treatment Number"
TRT01A = "Actual Treatment"
TRT01AN = "Actual Treatment Number"
REMISS = "Current Remission Status (Char)"
REMISSN = "Current Remission Status (Num)"
ITTFL = "Intent-To-Treat Population Flag"
SAFFL = "Safety Population Flag"
EFFL = "Efficacy-Evaluable Population Flag"
CA125FL = "Efficacy-Evaluable CA125 Population Flag"
TRTDCFL = "Treatment Discontinuation Flag"
COMPLFL = "Study Period Completers Flag"
STDDCFL = "Study Period Discontinuation Flag"
SFUDCFL = "Follow-Up Period Discontinuation Flag"
SFUFL = "Survival Follow-Up Period Entry Flag"
TRTSDT = "First Treatment Date (num)"
TRTEDT = "Last Treatment Date (num)"
FPDUR = "Safety Follow-Up Duration (Days)"
PARAM = "Analysis Parameter Description"
PARAMCD = "Analysis Parameter Short Name"
AVAL = "Analysis Value"
DTYPE = "Derivation Type";

```

```

keep STUDYID USUBJID SUBJID SITEID INVNAM INVID AGE AGEU AGEGR1 AGEGR1N
SEX RACE ETHNIC COUNTRY RANDDT TRTP TRT01P TRT01PN TRT01A TRT01AN
REMISS REMISSN ITTFL SAFFL EFFL CA125FL TRTDCFL COMPLFL STDDCFL
SFUDCFL SFUFL TRTSDT TRTEDT FPDUR PARAM PARAMCD AVAL DTYPE;

```

```
run;
```

```
libname vadex xport "&VALPATH/ADEX.xpt";
```

```
proc sort data=vadex.adex out=val_adex;
  by USUBJID PARAMCD;
run;
```

```
proc sort data=adam.adex;
  by USUBJID PARAMCD;
run;
```

```
data val_keep;
  set val_adex;
  keep USUBJID PARAMCD
  STUDYID SUBJID SITEID INVNAM INVID AGE AGEU AGEGR1 AGEGR1N SEX RACE ETHNIC COUNTRY
  RANDDT TRTP TRT01P TRT01PN TRT01A TRT01AN REMISS REMISSN ITTFL SAFFL EFFL CA125FL
  TRTDCFL COMPLFL STDDCFL SFUDCFL SFUFL TRTSDT TRTEDT FPDUR PARAM AVAL DTYPE;
run;
```

```
data adam.adex;
  merge adam.adex(in=a)
        val_keep(rename=(
          STUDYID=STUDYID_REF
          SUBJID=SUBJID_REF
          SITEID=SITEID_REF
          INVNAM=INVNAM_REF
          INVID=INVID_REF
          AGE=AGE_REF
          AGEU=AGEU_REF
          AGEGR1=AGEGR1_REF
          AGEGR1N=AGEGR1N_REF
          SEX=SEX_REF
          RACE=RACE_REF
          ETHNIC=ETHNIC_REF
          COUNTRY=COUNTRY_REF
          RANDDT=RANDDT_REF
          TRTP=TRTP_REF
          TRT01P=TRT01P_REF
          TRT01PN=TRT01PN_REF
          TRT01A=TRT01A_REF
          TRT01AN=TRT01AN_REF
          REMISS=REMISS_REF
          REMISSN=REMISSN_REF
          ITTFL=ITTFL_REF
          SAFFL=SAFFL_REF
          EFFL=EFFL_REF
          CA125FL=CA125FL_REF
          TRTDCFL=TRTDCFL_REF
          COMPLFL=COMPLFL_REF
          STDDCFL=STDDCFL_REF
          SFUDCFL=SFUDCFL_REF
          SFUFL=SFUFL_REF
          TRTSDT=TRTSDT_REF
          TRTEDT=TRTEDT_REF
          FPDUR=FPDUR_REF
          PARAM=PARAM_REF
          AVAL=AVAL_REF
          DTYPE=DTYPE_REF
        ));
  by USUBJID PARAMCD;
```

```

if a;

STUDYID  = STUDYID_REF;
SUBJID   = SUBJID_REF;
SITEID   = SITEID_REF;
INVNAM   = INVNAM_REF;
INVID    = INVID_REF;
AGE      = AGE_REF;
AGEU     = AGEU_REF;
AGEGR1   = AGEGR1_REF;
AGEGR1N  = AGEGR1N_REF;
SEX      = SEX_REF;
RACE     = RACE_REF;
ETHNIC   = ETHNIC_REF;
COUNTRY  = COUNTRY_REF;
RANDDT   = RANDDT_REF;
TRTP     = TRTP_REF;
TRT01P   = TRT01P_REF;
TRT01PN  = TRT01PN_REF;
TRT01A   = TRT01A_REF;
TRT01AN  = TRT01AN_REF;
REMISS   = REMISS_REF;
REMISSN  = REMISSN_REF;
ITTFL    = ITTFL_REF;
SAFFL    = SAFFL_REF;
EFFL     = EFFL_REF;
CA125FL  = CA125FL_REF;
TRTDCFL  = TRTDCFL_REF;
COMPLFL  = COMPLFL_REF;
STDDCFL  = STDDCFL_REF;
SFUDCFL  = SFUDCFL_REF;
SFUFL    = SFUFL_REF;
TRTSDT   = TRTSDT_REF;
TRTEDT   = TRTEDT_REF;
FPDUR    = FPDUR_REF;
PARAM    = PARAM_REF;
AVAL     = AVAL_REF;
DTYPE    = DTYPE_REF;

drop STUDYID_REF SUBJID_REF SITEID_REF INVNAM_REF INVID_REF AGE_REF AGEU_REF
      AGEGR1_REF AGEGR1N_REF SEX_REF RACE_REF ETHNIC_REF COUNTRY_REF RANDDT_REF
      TRTP_REF TRT01P_REF TRT01PN_REF TRT01A_REF TRT01AN_REF REMISS_REF REMISSN_REF
      ITTFL_REF SAFFL_REF EFFL_REF CA125FL_REF TRTDCFL_REF COMPLFL_REF STDDCFL_REF
      SFUDCFL_REF SFUFL_REF TRTSDT_REF TRTEDT_REF FPDUR_REF PARAM_REF AVAL_REF DTYPE_REF;

run;

.....

proc sort data=adam.adex;
  by USUBJID PARAMCD;
run;

libname xadex xport "&OUTPATH/ADEX.xpt";

.....

proc copy in=adam out=xadex;
  select adex;
run;

```

```
proc sort data=vadex.adex out=base_adex;  
  by USUBJID PARAMCD;  
run;
```

```
proc sort data=adam.adex out=comp_adex;  
  by USUBJID PARAMCD;  
run;
```

```
ods rtf file="&OUTPATH/ADEX_PROC_COMPARE.rtf";
```

```
title "PROC COMPARE: Validation ADEX vs Created ADEX";
```